AYE
TECH HUB

• PROTASTRUCTURE • LESSON 01

Introduction to ProtaStructure

Structural Analysis, Design and Detailing Software

Lesson 01 of 30 | ProtaStructure Program

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- What ProtaStructure is and who uses it globally
- The complete 6-stage structural design workflow in ProtaStructure
- Supported codes: Eurocode, ACI 318, BS 8110, ASCE 7 and more
- Structural element types: beams, columns, slabs, walls, foundations
- ProtaStructure vs ETABS, SAP2000, SAFE and STAAD.Pro

What Is ProtaStructure?

ProtaStructure is a **fully integrated structural engineering software** for the analysis, design, detailing, and documentation of reinforced concrete, steel, and composite structures. Developed by Prota Software (Turkey), it is used by structural engineers across 100+ countries for residential, commercial, industrial, and infrastructure projects.

What makes ProtaStructure unique is its **all-in-one workflow**: you model the structure, run the analysis, perform code-compliant design checks, produce reinforcement detailing, generate bar schedules, and export calculation reports — all within a single integrated environment. There is no need to transfer data between separate programs for analysis and design, which eliminates transcription errors.

The 6-Stage ProtaStructure Workflow

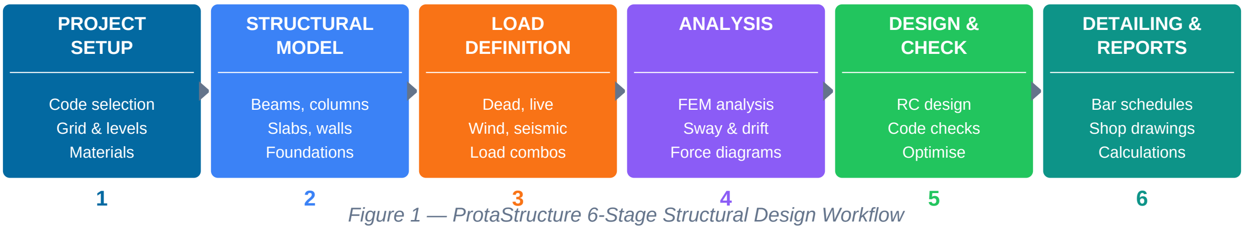


Figure 1 — ProtaStructure 6-Stage Structural Design Workflow

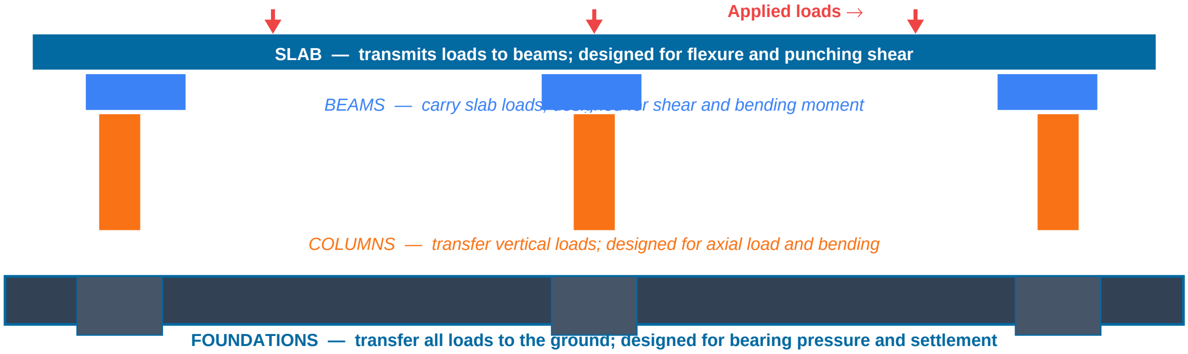
Stage	Key Actions	ProtaStructure Feature
1 — Project Setup	Select design codes and load combinations; set grid and floor levels; define material grades	Code library (EC2, ACI, BS); material database; grid editor
2 — Structural Model	Input beams, columns, slabs, walls, and foundations using BIM-style 3D entry	Graphical 3D model editor; DXF/DWG import; Revit link
3 — Load Definition	Apply dead loads (self-weight + superimposed), live loads, wind, seismic, and snow	Automatic self-weight; load generator; seismic wizard
4 — Analysis	Run finite element analysis; check sway, drift, and deformation; view force diagrams	FEM solver; 3D analysis results viewer; modal analysis
5 — Design and Code Check	Automatic RC design for all elements; code ratio checks; manual optimisation	EC2/ACI code checks; design optimiser; detailed design tables
6 — Detailing and Reports	Generate bar bending schedules, reinforcement drawings, and calculation reports	Automatic detailing; ProtaDetails export; PDF/DXF reports

BIM integration: ProtaStructure links directly to Revit Structure via a two-way IFC/direct link. Architectural and MEP models can be imported as reference models for coordination. Structural model data can be exported to Revit, enabling a fully coordinated BIM workflow between architect, structural engineer, and MEP engineer.

Reinforced Concrete Structural Systems

Before modelling in ProtaStructure, you must understand the **structural system** — how gravity and lateral loads are transferred from the top of the building down to the foundations. ProtaStructure handles all the standard RC structural systems.

Figure 2 — Reinforced Concrete Structural System: Load Path



Structural Element	Function in Building	ProtaStructure Design Checks
Slab (flat/ribbed/waffle)	Carries gravity loads (dead + live) and transfers to beams or columns/walls	Flexure, shear, punching shear, deflection, minimum reinforcement
Beam (rectangular/T/L)	Receives slab loads and transfers to columns; provides lateral stiffness in frames	Flexure (hogging + sagging), shear, torsion, deflection, crack width
Column	Transfers axial loads and bending moments from beams down to foundations	Axial + biaxial bending interaction, slenderness, confinement, bar arrangement
Shear Wall / Core Wall	Provides lateral resistance to wind and seismic forces; stabilises the building	In-plane shear, flexural design, boundary element design (seismic)
Pad Footing / Strip Foundation	Spreads column/wall loads to the soil below the required bearing pressure	Bearing pressure check, punching, flexure, sliding, overturning
Raft Foundation	Covers entire building footprint; distributes loads to poor ground uniformly	Soil-structure interaction, differential settlement, thickened edges

Supported Design Codes

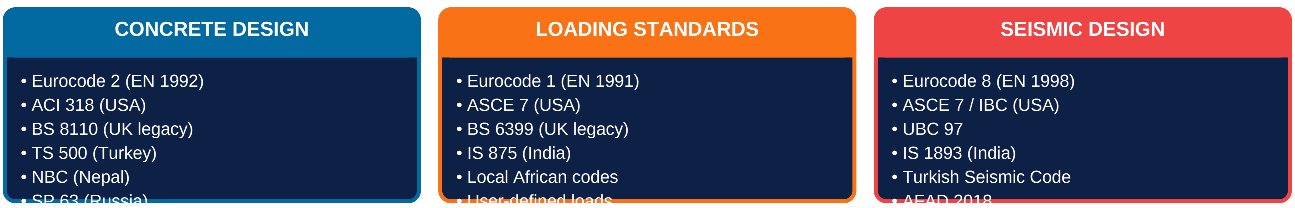


Figure 3 — ProtaStructure Supported Design Codes

Which code to use in Africa? Ethiopia: EBCS (Ethiopian Building Code Standard) + EC2 base. Kenya, Tanzania, Uganda: British Standards (BS 8110 / BS 6399) are widely used. South Africa: SANS 10160 loading + SANS 10100 concrete. Egypt, Libya, Morocco: Egyptian Code (ECP) or Eurocode. Most international contractors use Eurocode regardless of country. ProtaStructure supports all of these.

ProtaStructure vs Other Structural Software

Software	Best For	Strengths	Limitations
ProtaStructure	RC buildings — residential, commercial, multi-storey	All-in-one: model → analysis → design → detailing. Fast workflow. Multi-code.	Less common in North America; fewer plugins than ETABS
ETABS (CSI)	Tall buildings, complex lateral analysis, performance-based design	Industry gold standard for high-rise; advanced seismic analysis; very powerful	No integrated detailing; requires SAFE or separate tools for RC detailing
SAP2000 (CSI)	Bridges, industrial structures, non-standard geometry, research	Most versatile FEM solver; handles almost any structure type	Steep learning curve; no automatic RC detailing; general purpose
SAFE (CSI)	Slab and foundation design specifically	Best-in-class slab and mat foundation design	Not a full building analysis tool; requires ETABS for superstructure
STAAD.Pro (Bentley)	Steel structures, industrial buildings, oil & gas	Strong steel design; widely used in India and Middle East	RC workflow less intuitive than ProtaStructure or ETABS
Tekla Structures	Steel fabrication detailing and BIM	Best steel connection design and fabrication shop drawings	Not a structural analysis tool; focused on detailing only

System Requirements for ProtaStructure 2024/2025

Component	Minimum	Recommended
OS	Windows 10 64-bit	Windows 11 64-bit
CPU	Intel/AMD, 4 cores, 2.5 GHz	8 cores+, 3.5 GHz (analysis speed scales with cores)
RAM	8 GB	32 GB (essential for large multi-storey buildings)
GPU	2 GB VRAM, OpenGL 3.3+	8 GB VRAM (better 3D viewport performance)
Storage	10 GB free (SSD)	NVMe SSD — analysis temp files are large
Display	1280×1024	1920×1080+ dual monitors (model + results simultaneously)

Career Pathways for ProtaStructure Engineers

Role	Core Skills	Typical Employer	Demand in Africa
Junior Structural Engineer	ProtaStructure modelling, basic RC design, drawing production	Consulting firms, construction companies	High — most concrete building projects need this
Structural Design Engineer	Full analysis, multi-storey buildings, code compliance	Structural consultancies, international contractors	Very High — qualified engineers are scarce
BIM Structural Specialist	Revit Structure + ProtaStructure + Navisworks coordination	International contractors, large design firms	High and growing — BIM mandates expanding

Freelance Structural Designer	ProtaStructure + AutoCAD for independent project delivery	Self-employed, small developers	High — structural drawings required for all buildings

What comes next — PROTA-002: ProtaStructure Interface and Navigation — a complete tour of every panel, toolbar, and menu. You will navigate a project, understand the 3D viewer, and set up your first project template ready for modelling.